MG-PE-04 PE Node Failure SOP - Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: MG-PE-04 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_2873

Description

The MG-PE-04 PE node failure event is precipitated by a hardware/environmental fault, specifically ASIC memory parity errors, line card crashes, or power supply redundancy loss. This failure domain is inherent to the PE role and Cisco vendor equipment due to the complex interplay of packet forwarding, line card management, and redundancy protocols. The PE node's primary function is to perform packet forwarding, which involves scrutinizing incoming packets at the ingress interface, processing them, and forwarding them to the destined egress interface. The PE node's line cards are responsible for handling the ingress and egress traffic, and any failure within these line cards can lead to packet forwarding errors and subsequent ASIC memory parity errors. Furthermore, the PE node's power supply redundancy is critical to maintaining uptime and preventing line card crashes. In the event of a hardware or environmental fault, the PE node will generate an ALARM_2873 alarm, indicating a warning severity.

Pre-requisite Checks

- Verify that the MG Road Core site is accessible and the remote hands are dispatched to perform the necessary tasks.

- Check the redundant link status to ensure that it is active and functioning correctly.

- Perform a physical inspection of the MG-PE-04 PE node to identify any visible signs of hardware failure.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers cpu-usage, Expected Output: CPU usage information for the PE node, Description: This command is used to diagnose CPU usage patterns, which can help identify potential bottlenecks or issues related to packet forwarding.
```

```
Command: show interfaces gigabitethernet 0/0/0, Expected Output: Interface statistics for the ingress interface, Description: This command is used to diagnose interface statistics, which can help identify potential issues related to packet forwarding or line card management.

Command: show redundancy, Expected Output: Redundancy information for the PE node, Description: This command is used to diagnose redundancy protocols, which can help identify potential issues related to power supply redundancy or line card crashes.
```

Resolution Steps

Step 1: Step: Replace the faulty line card, Command: no controllers cpu-usage, Description: This command is used to disable CPU usage monitoring, which can help prevent further packet forwarding errors.

Step 2: Step: Reboot the PE node, Command: reload, Description: This command is used to reboot the PE node, which can help resolve issues related to line card crashes or power supply redundancy loss.

Step 3: Step: Verify power supply redundancy, Command: show redundancy, Description: This command is used to verify power supply redundancy, which can help identify potential issues related to power supply redundancy loss.

Post-Check Verification

```
Command: show controllers cpu-usage, Expected Output: CPU usage information for the PE node, Description: This command is used to verify CPU usage patterns, which can help confirm that the issue has been resolved.

Command: show interfaces gigabitethernet 0/0/0, Expected Output: Interface statistics for the ingress interface, Description: This command is used to verify interface statistics, which can help confirm that the issue has been resolved.

Command: show redundancy, Expected Output: Redundancy information for the PE node, Description: This command is used to verify redundancy protocols, which can help confirm that the issue has been resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: This command is used to restore the original configuration, which can help roll back changes made during the resolution process.

- Step: Revert the faulty line card, Command: no controllers cpu-usage, Description: This command is used to revert the faulty line card, which can help roll back changes made during the resolution process.